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Investigating the effect of systematically modifying the molar ratio of hydrogen bond donor and acceptor on solvation characteristics of deep eutectic solvents formed using choline chloride salt and polyalcohols

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ABSTRACT

Choline chloride-based deep eutectic solvents (DESs) are immensely popular in organic synthesis, catalysis, electrochemistry, and separation science. A popular choice of hydrogen bond donor (HBD) among these DESs consists of both straight-chain and branched polyols that can incorporate additional functional groups, such as ether linkages. Previous studies have shown that the extraction efficiency is significantly altered when the molar ratio of HBD in choline chloride-based DES systems is varied, but no study has been able to relate it to their solvation characteristics. This is largely due to the limited sensitivity of existing solvatochromic dye techniques to detect minor changes in solvation interactions when the DES composition is varied. In this study, inverse gas chromatography was employed for the first time to investigate the variation in solvation properties for DESs comprised of choline salts as hydrogen bond acceptors (HBAs) and polyols as HBDs when their HBA/HBD ratio is systematically altered. Unlike many organic solvents, DES systems investigated in this work possessed a significant hydrogen bond character. It was observed that the hydrogen bond basicity generally plateaued at higher molar ratios of HBD while the hydrogen bond acidity was observed to be the highest at HBA/HBD ratios of 1:10 in all DESs. Amongst all solvents, neat HBDs (triethylene glycol and 1,8-octane diol) possessed the weakest hydrogen bond basicity since they lack the chloride anion that acts as the primary hydrogen bond acceptor. Results from this study demonstrate that the solvation characteristics of DESs are largely different from their starting materials while the HBA/HBD ratio further influences their solvation interactions that can in turn impact important parameters such as extraction yields.

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1. Introduction

Deep eutectic solvents (DESs) have become an increasingly popular choice of solvents in scientific research [1]. DESs are two-component systems formed by mixing precise quantities of a hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA), both of which often possess very high melting points [2,3]. The resulting homogenous mixture typically melts below room temperature. Most HBAs are comprised of ammonium, phosphonium, and imidazolium salts while the choice of HBD generally consists of organic acids, amides, alcohols, and carbohydrates [3,4]. DESs possess notable advantages over conventional solvents in catalysis [5,6], organic synthesis [7,8], separations [9,10], and electrochemical applications [11] as they often possess desirable properties such as low vapor pressure and tunable viscosities, and are often cheaper to prepare [12,13].

Choline chloride ($[Ch^+][Cl^-]$) and its derivatives are commonly employed as HBAs to form popular and highly versatile DESs [14]. Compared to tetraalkylammonium/phosphonium salts, choline chloride is relatively biodegradable, possesses lower toxicity, and can form DESs with a wider variety of HBDs [2,9,15]. Additionally, the solubilization of DESs in small quantities of water can prove useful in regulating their fluidity, conductivity, and polarity [16]. Depending on the HBD, DESs often possess both hydrophilic and moderately hydrophobic functional groups that can simultaneously interact with both polar and non-polar analytes. Clearly, choline chloride-based DESs possess higher structural tunability than many other classes of DESs, thereby enhancing their designer solvent properties [14,17].

Several studies have reported that the physico-chemical properties of DESs are dependent on the choice of HBA or HBD [17,18]. For example, natural deep eutectic solvents (NADES) comprised of the $[Ch^+][Cl^-]$ HBA were reported to be less dense than the ones formed using non-ionic components while those containing polar and organic acid-based HBDs were found to possess higher con-

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ductivities [19]. Additionally, the stoichiometric ratio between the HBA and HBD serves as a tool to further modify specific physical properties as DESs composed of a higher HBD molar ratio generally tend to possess lower viscosities [20], while a change in HBA/HBD ratio impacts their melting point [21,22] and ultimately influences the temperature at which they can be employed as solvents. Clearly, varying the type of HBA and HBD and/or their ratio can simultaneously influence multiple physical and chemical properties of DESs. While trends in surface tension, melting point, conductivity, and density have been thoroughly studied [16,21,23], very little is understood concerning the solvation characteristics of DESs [17].

In an attempt to further understand the solvation properties of DES systems, various studies have investigated the solvatochromic response of certain dyes in the presence of DESs to construct the normalized polarity scale ($E_T N$) [24,25]. For example, $E_T N$ values of 0.35 and 0.84 were obtained for the [Ch^+][Cl^-]: 2 glycerol and [Ch^+][Cl^-]: 2 levulinic acid DESs, respectively, demonstrating a significant difference in their solvation strength [25]. In contrast to the $E_T N$ scale, Kamlet-Taft solvent parameters measure the solvent's hydrogen bond donating ability (α), hydrogen bond accepting ability (β), and dipolarity/polarizability (π^*) by employing several solvatochromic dyes [25,26]. While the solvatochromic approach has contributed immensely to our knowledge of DES solvation characteristics, some cases are still plagued with ambiguity. Pandey et al. reported $E_T N$ values of 0.84 for both glycerol and the [Ch^+][Cl^-]: 2 glycerol DES, effectively showing no difference in polarity between the DES and pure HBD [27]. However, large differences in total phenolic content (TPC) of *Chlorella vulgaris* were measured when the [Ch^+][Cl^-]: 2 glycerol DES (2.5 mg gallic acid equivalent [GAE]/g of biomass) was employed as an extraction solvent compared to neat glycerol (0.43 mg GAE/g of biomass) [28]. In the case of orange peel waste, a TPC of over 5 mg GAE/g of biomass was obtained using the [Ch^+][Cl^-]: 2 ethylene glycol DES compared to under 1.5 mg GAE/g of biomass determined using pure ethylene glycol [29], even though a $E_T N$ value of 0.82 was reported for both the DES and neat HBD [27]. Abbott and co-workers studied the 1:1, 1:1.5, 1:2, and 1:3 molar ratios of [Ch^+][Cl^-] and glycerol DES using the Kamlet-Taft approach and reported no difference in their hydrogen bond accepting ability (values ranged from 0.657 to 0.658) [30], even though the measured TPC of orange peel waste using these eutectic mixtures varied from 1.37 to 2.90 mg GAE/g of biomass [29]. The authors concluded that the solvatochromic approach is insensitive to compositional changes in high polarity solvents such as DESs, thereby limiting its potential in explaining their solvation characteristics [30].

Given the limitations of the solvatochromic approach, an alternative technique is needed to investigate small variations in DES solvation characteristics. Inverse gas chromatography (IGC) is a sensitive technique that measures solute-solvent interactions by studying the chromatographic retention of probe molecules to a DES stationary phase [31,32]. The probe molecules exhibit multiple types of unique and simultaneous interactions with the DESs and can be quantified using a linear free energy relationship [33]. While choline chloride-based DESs comprised of different classes of HBD have been previously studied using IGC [34], the effect of varying their HBA/HBD ratio has not been thoroughly investigated. Most studies involving catalysis, organic synthesis, and separations often screen multiple combinations of the same HBA and HBD in order to identify the optimal eutectic mixture. Thus, it is vital to understand the impact on DES solvation interactions when the HBA/HBD ratio is varied.

In this study, three different DES systems that are popularly employed in separation studies and comprised of choline chloride and acetate salts as HBAs and triethylene glycol (TEG) and 1,8-octane diol (1,8-OD) as HBDs were investigated using IGC. The impact of an incremental change in molar ratio of HBD on DES solvation interactions was studied by employing 6 different molar combinations in each DES system. Additionally, the solvation characteristics of DESs were compared with neat HBDs that were employed in preparing them. By choosing multiple eutectic mixtures in each DES system, it was established that their solvation interactions can vary significantly with minor changes in HBA/HBD ratio, irrespective of the type of choline salt or HBD employed to form the DESs. In addition, the anion and cation components were observed to result in vastly different solvation properties compared to the neat HBDs. This study demonstrates that a particular molar composition of DES offers a unique solvation character that is largely different from that of its starting materials. The resulting solvation characteristics can significantly influence its efficacy as a catalyst or reaction medium, thereby making it essential to use these solvation properties as the basis to identify the optimal eutectic mixture for a given catalytic and synthetic route.

2. Experimental

2.1. Materials, reagents, DES preparation and characterization

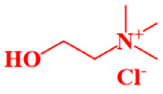
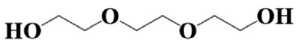
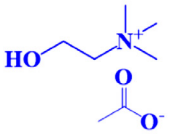
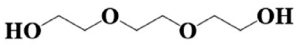
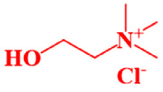
All reagents employed in this study are provided in the supporting information. The reagents and starting materials employed in DES preparation were extensively dried under vacuum for 48 h at 25 °C and stored continuously in a desiccator. All DESs were prepared using the grinding method without use of heat. Desired quantities of choline salts and HBDs (see Table 1) were carefully transferred into glass vials using a clean spatula for solid materials and a transfer pipette for liquid reagents. The contents of each vial were ground together until a homogenous liquid was obtained. The purity of all DESs was characterized using nuclear magnetic resonance (NMR) spectroscopy; the ^{13}C and ^{1}H NMR spectra are provided in the supporting information. The water content of each DES was measured in triplicate using a Metrohm 831 Karl Fischer coulometric titrator. All DESs had water content under 0.8 wt.%, as shown in Table S1.

The melting points of all DESs were studied using a DSC Q2000 calorimeter (TA Instruments). DESs that appeared as liquids at room temperature were cooled from 40 °C to -120 °C at a rate of 20 °C min^{-1} and subsequently heated to 80 °C at the same rate and DESs appearing as solids under ambient conditions were cooled to -65 °C at a rate of 20 °C min^{-1} before being heated to 200 °C at the same rate. Data obtained from DSC measurements were analyzed using TA universal analysis software (TA Instruments) and the melting point and/or glass transition temperatures for all DESs are provided in Table S2.

2.2. Preparation of chromatographic columns and characterization method

DESs were employed as stationary phases in the preparation of chromatographic columns using the static method of column coating. A volatile solvent system comprised of dichloromethane containing 5% methanol (by volume) was employed to prepare DES coating solutions having a concentration of 0.32% w/v. In addition, TEG and 1,8-OD were dissolved in pure dichloromethane to prepare coating solutions of neat HBDs, as well. Deactivated fused silica capillary was cut into 5 m segments which were filled with the coating solutions of DESs or neat HBD. One end of each capillary was sealed while the other end was connected to a vacuum before placing it into a thermally-regulated water bath maintained at 40 °C. The solvent was evaporated leaving behind a thin layer of DES or neat HBD on the inner capillary wall having an approximate film thickness of 0.20 µm [35]. All chromatographic columns

Table 1
Chemical structures and numbering scheme used for three deep eutectic solvent systems examined in this study.

DES system	HBA	HBD	Ratio	DES abbreviation
1			1:2	[Ch⁺][Cl⁻] : 2 TEG
			1:3	[Ch⁺][Cl⁻] : 3 TEG
			1:4	[Ch⁺][Cl⁻] : 4 TEG
			1:5	[Ch⁺][Cl⁻] : 5 TEG
			1:6	[Ch⁺][Cl⁻] : 6 TEG
	[Ch⁺][Cl⁻]	TEG	1:10	[Ch⁺][Cl⁻] : 10 TEG
2			1:2	[Ch⁺][Ace⁻] : 2 TEG
			1:3	[Ch⁺][Ace⁻] : 3 TEG
			1:4	[Ch⁺][Ace⁻] : 4 TEG
			1:5	[Ch⁺][Ace⁻] : 5 TEG
			1:6	[Ch⁺][Ace⁻] : 6 TEG
	[Ch⁺][Ace⁻]	TEG	1:10	[Ch⁺][Ace⁻] : 10 TEG
3			1:2	[Ch⁺][Cl⁻] : 2 (1,8-OD)
			1:3	[Ch⁺][Cl⁻] : 3 (1,8-OD)
			1:4	[Ch⁺][Cl⁻] : 4 (1,8-OD)
			1:5	[Ch⁺][Cl⁻] : 5 (1,8-OD)
			1:6	[Ch⁺][Cl⁻] : 6 (1,8-OD)
	[Ch⁺][Cl⁻]	1,8-OD	1:10	[Ch⁺][Cl⁻] : 10 (1,8-OD)

were installed in a gas chromatograph equipped with a flame ionization detector and conditioned at 60 °C for 25 min under a constant flow of helium (1 mL/min). Column efficiencies ranged from 1450 to 3850 plates/meter, as determined by separating a naphthalene solution (1 mg/mL) at 60 °C.

Chromatographic columns of DESs and neat HBDs were characterized using Agilent 6890 N and 7890 B gas chromatographs (Santa Clara, California). The oven temperature was varied between 30 and 70 °C while the inlet/detectors were maintained at 150 °C. Additionally, the flow of helium through the chromatographic columns was kept constant at 1 mL/min while the inlet split ratio was maintained at 20:1. Probe molecules were dissolved in dichloromethane at a concentration of 1 mg/mL and 1 µL of each standard solution was injected for separation. Table S3 provides a comprehensive list of 50 probe molecules employed in this study. Multiple linear regression analysis (MLRA) and statistical calculations were performed using Analyze-it (Microsoft, USA).

3. Results and discussion

The versatility of choline chloride-based DESs enables a given HBA and HBD to combine and form multiple eutectic mixtures that differ in their HBA/HBD ratio. For example, six HBA/HBD ratios (i.e., 1:1, 1:2, 1:3, 1:4, 1:5, and 1:6) have been reported for the DES system formed using the [Ch⁺][Cl⁻] salt as HBA and 1,3-butane diol as HBD [36]. While these eutectic mixtures may not qualify as DESs due to their significant deviation from the theoretical eutectic point, the term “pseudo-DES mixtures” has been used to describe similar systems [37]. In extraction studies that employ choline chloride-based DESs comprised of polyols as HBDs, all molar combinations of a DES system are often screened to identify and employ the eutectic mixture that results in optimal extraction efficiencies [28,36,38]. For example, Yao and co-workers employed all 6 eutectic mixtures (molar ratios of 1:1, 1:2, 1:3, 1:4, 1:5, and 1:6) of the DES comprised of [Ch⁺][Cl⁻] as the HBA and 1,3-butane diol as HBD in extracting phenolic compounds from *Pyrola incarnata Fisch* [36]. The authors reported an increase in extraction yields of hyperin (from 1.1898 ± 0.0369 to 1.5206 ± 0.0304 mg/g) in employing the 1:1 to 1:5 molar ratio of HBD and a subsequent decrease to 1.3886 ± 0.0361 mg/g at a HBA/HBD ratio of 1:6 [36]. It was reported elsewhere that the increase in extraction yields using higher molar ratios of HBD can be attributed to a decrease in DES viscosity, thereby enhancing the amount of analyte extracted [39]. However, at a HBA/HBD ratio of 1:6, the decrease in [Ch⁺][Cl⁻] content results in reduced interactions between phenolics and the chloride anion. While the decrease in DES viscosity with an increase in molar ratio of HBD has been extensively reported and supports this claim [36,39], no study currently exists that demonstrates this effect at HBA/HBD ratios of 1:6 or higher. Therefore, this study explores the variations in solvation interactions for different molar combinations of DES systems comprised of choline salts as HBAs and polyols as HBDs when small changes are made to their HBA/HBD ratio.

The DES systems evaluated in this study were carefully chosen to reflect those that can form multiple eutectic mixtures and are extensively employed in separations, catalysis, and organic synthesis. The [Ch⁺][Cl⁻] salt was coupled with TEG and 1,8-OD (DES systems 1 and 3 in Table 1) to demonstrate the effect of systematically varying the HBA/HBD ratio on solvation interactions in DES combinations that possess a common HBA. In DES system 3, 1,8-OD was chosen to study the effect of eliminating the ether functional groups from TEG. Additionally, choline acetate ([Ch⁺][Ace⁻]) was paired with TEG (DES system 2) to examine whether or not variations in DES solvation interactions are limited to a specific anion in the HBA when the molar ratio of HBD is varied. Furthermore, neat HBDs (TEG and 1,8-OD) used to prepare all 3 DES systems were also investigated to compare their solvation characteristics with those of their corresponding DESs. The chromatographic retention of probe molecules on DES systems 1 and 2 was obtained for temperatures ranging between 30 and 60 °C. However, eutectic mixtures of DES system 3 comprised of [Ch⁺][Cl⁻] as the HBA and 1,8-OD as HBD were solids at temperatures lower than 60 °C. Therefore, DES system 3 was studied at temperatures of 60 and 70 °C.

To relate the chromatographic retention of analytes to individual solute-solvent interactions, the Abraham solvation parameter

model (Eq. (1)) was employed [33].

$$Log\ k = c + eE + sS + aA + bB + lL \quad (1)$$

In Eq. (1), k represents the retention factor of each probe molecule calculated using the dead time and the retention time of each chromatographic column. The solute descriptors (*E, S, A, B,* and *L*) are specific to analytes and represent the excess molar refraction, dipolarity/polarizability, hydrogen bond acidity, hydrogen bond basicity, and gas-hexadecane partition coefficient at 298 K, respectively [33]. The system constants (*e, s, a, b,* and *l*) represent specific solvation interactions of the DES stationary phase, where *e* measures n-π and π-π interactions, *s* the dipolarity/polarizability, *a* the hydrogen bond basicity, *b* the hydrogen bond acidity, and *l* provides a measure of dispersive-type interactions [40].

3.1. Effect of varying the HBA/HBD ratio on system constants and chromatographic retention of analytes

To explore the effect of systematically modifying the HBA/HBD ratio, each DES system was thoroughly investigated to identify changes in its solvation characteristics. As hypothesized, the chromatographic retention of analytes were dramatically altered when the molar ratio of HBD was varied. Additionally, the system constants representing individual solvation interactions were significantly influenced by changes in HBA/HBD ratio. Table 2 provides the system constants for neat TEG and DES systems 1 and 2 formed by pairing with the [Ch$^+$][Cl$^-$] and [Ch$^+$][Ace$^-$] HBAs, respectively. It can be observed that most system constants for TEG and the corresponding DESs increased significantly when the temperature was decreased. A similar trend was observed in the system constants for neat 1,8-OD and its DESs comprised of [Ch$^+$][Cl$^-$] as the HBA (DES system 3), as shown in Table 3. The models developed for each DES system were statistically sound based on the value of the Fisher *F*-statistics (generally higher than 800) as well as high correlation coefficients (0.99) of the multiple linear regression fit. Each DES system exhibited significant dipolarity (*s*-term), hydrogen bond basicity (*a*-term), hydrogen bond acidity (*b*-term), and dispersive-type interactions (*l*-term). Dipolarity was observed to generally decrease upon increasing the molar ratio of HBD with values ranging from 2.17 to 2.33 for DES system 1, 2.01 to 2.39 for DES system 2, and 1.50 to 1.84 for DES system 3 at 60 °C. In contrast, dispersion interactions generally increased from 0.43 to 0.51 for DES system 1, 0.50 to 0.58 for DES system 2, and 0.64 to 0.69 for DES system 3 when the molar ratio of HBD was increased.

The hydrogen bonding component in these 3 DES systems experienced the greatest change when the HBA/HBD ratio was varied. In DES system 1, the hydrogen bond basicity (*a*-term) slightly decreased from (4.39 ± 0.13) to (4.21 ± 0.11) at 60 °C (Table 2) for the [Ch$^+$][Cl$^-$]: 2 TEG and [Ch$^+$][Cl$^-$]: 3 TEG DESs, respectively. For the same molar ratios of HBD, the decrease from (4.68 ± 0.12) to (4.39 ± 0.10) in DES system 2 and (4.50 ± 0.13) to (4.18 ± 0.09) in DES system 3 (Table 3) was more drastic, demonstrating that the hydrogen bond basicity can be significantly changed when the HBA/HBD ratio is varied only slightly. However, at higher molar ratios of HBD, the hydrogen bond basicity generally plateaued for all 3 DES systems. In contrast, stronger hydrogen bond acidity interactions were observed at high molar ratios of HBD. For example, in DES system 3, the *b*-term slightly increased from (0.32 ± 0.08) to (0.40 ± 0.08) at 60 °C for the [Ch$^+$][Cl$^-$]: 3 (1,8-OD) and [Ch$^+$][Cl$^-$]: 5 (1,8-OD) DESs, respectively, with the highest value of (0.48 ± 0.10) for the [Ch$^+$][Cl$^-$]: 10 (1,8-OD) DES. These trends were further supported by variations in the chromatographic retention of basic probe molecules, as shown in Table 4. For example, retention factors of *N,N*-dimethylacetamide (*N,N*-DMAC) at 60 °C consistently increased from 39.36 to 62.72 in DES system 3, indicating a stronger affinity for basic probe molecules at higher molar ratios of HBD. A similar trend was observed for DES system 1 and 2 as well, demonstrating that the strength of interaction with basic probe molecules is further enhanced, irrespective of the choice of choline salt or HBD forming these DESs.

To further demonstrate the effect of varying the HBA/HBD ratio on the chromatographic retention of analytes, a mixture of 16 probe molecules was separated on 5 different molar combinations of DES system 2 (HBA/HBD ratios of 1:2, 1:3, 1:4, 1:6, and 1:10) at 50 °C (see Fig. 1A–E). Additionally, the same probes were also subjected to chromatographic separation on a stationary phase composed of neat TEG (Fig. 1F) to compare the variation in analyte retention to that of its DESs. It can be observed that the retention of most probe molecules was the lowest on the [Ch$^+$][Ace$^-$]: 2 TEG DES and generally highest at larger molar ratios of HBD. Alcohols (acidic analytes) and basic probe molecules (*N,N*-DMF and *N,N*-DMAC) exhibited varying retention across all molar combinations of DES system 2. For example, 1-hexanol and cyclohexanol (analytes 11 and 14) possessed retention times under 6 and 10 min, respectively, on the [Ch$^+$][Ace$^-$]: 2 TEG DES (Fig. 1A) compared to over 10 and 12.5 min on the [Ch$^+$][Ace$^-$]: 4 TEG DES (Fig. 1C). The retention of these acidic probes either plateaued or slightly decreased when the molar ratio of HBD was further increased (Fig. 1D and E). In contrast, *N,N*-DMAC (analyte 13) retained the lowest (under 9 min) on the [Ch$^+$][Ace$^-$]: 2 TEG DES and the highest (over 14 min) on the [Ch$^+$][Ace$^-$]: 10 TEG DES. These results demonstrate that alcohols and other acidic analytes generally exhibit the highest retention on the 1:4 molar ratio of DES system 2. In addition, it was observed that the analyte elution order for DESs comprised of lower molar ratios of HBD (i.e., 1:2 and 1:3) was significantly different from those composed of higher molar ratios (i.e., 1:6 and 1:10). The retention factors of various analytes, provided in Table 4, demonstrate that these trends are also valid for DES systems 1 and 3 at 60 °C.

Using Kamlet-Taft parameters, Abbott et al. previously investigated the solvation characteristics for the 1:1, 1:2 and 1:3 molar ratios of DESs comprised of [Ch$^+$][Cl$^-$] as the HBA and glycerol as HBD, reporting no significant difference in hydrogen bond donating ability (α) values of 0.923, 0.937, and 0.914 and hydrogen bond accepting ability (β) values of 0.658, 0.657, and 0.657, respectively [30]. It was demonstrated that the solvatochromic method suffers a deficiency in characterizing the Lewis basicity of polar solvents such as DESs as solute-solvent interactions are dominated by solvent-solvent interactions, thereby making the solvatochromic indicators insensitive to small changes in DES composition. While they acknowledged the limitations of the solvatochromic approach in studying DESs [30], it has been reported elsewhere that Kamlet-Taft parameters do not vary significantly within the same class of solvents [25]. However, the chromatographic approach in this study does not possess such limitations as the hydrogen bond basicity, in addition to other solvation interactions, was found to vary for all 3 DES systems at identical molar ratios of HBD.

Several extraction studies that employ DESs comprised of [Ch$^+$][Cl$^-$] as the HBA and polyols as HBD often use them to extract phenolic compounds from a diverse variety of matrices [28,29,36,39]. Under an optimized set of conditions, the optimal HBA/HBD ratio of each DES system is often reported. In most of these studies, the 1:4 molar ratio of HBD has been extensively reported as the optimal eutectic mixture [36,38]. Amongst all of the probe molecules employed in this work, alcohols possess similar substituents (i.e., hydroxyl functional group) and closely resemble phenolic compounds. The chromatographic retention data for alcohols obtained from Table 4 and Fig. 1 demonstrates that these analytes generally retain the most on the 1:4 molar ratio of each DES system. Therefore, these results support the previous claim that the decrease in [Ch$^+$][Cl$^-$] content of a DES at higher molar

Table 2
System constants for triethylene glycol and deep eutectic solvents comprised of choline chloride and choline acetate salts as hydrogen bond acceptor and triethylene glycol hydrogen bond donor.

No.	DES/neat HBD	Temp. (°C)	System constants								
			c	*e*	*s*	*a*	*b*	*l*	*n* [a]	R^2 [b]	*F* [c]
1	[Ch^+][Cl^-]: 2 TEG	30	-3.29 (0.07)	0.26 (0.08)	2.54 (0.09)	5.08 (0.16)	0 (0)	0.54 (0.02)	37	0.99	757
		40	-3.25 (0.06)	0.28 (0.06)	2.38 (0.08)	4.72 (0.13)	0 (0)	0.50 (0.02)	37	0.99	967
		50	-3.19 (0.06)	0.27 (0.07)	2.34 (0.08)	4.43 (0.13)	0 (0)	0.45 (0.02)	37	0.99	833
		60	-3.35 (0.06)	0.27 (0.06)	2.33 (0.08)	4.39 (0.13)	0 (0)	0.43 (0.02)	37	0.99	860
		30	-3.36 (0.07)	0.19 (0.07)	2.55 (0.08)	5.00 (0.13)	0 (0)	0.57 (0.02)	36	0.99	1019
2	[Ch^+][Cl^-]: 3 TEG	40	-3.37 (0.06)	0.19 (0.06)	2.47 (0.07)	4.73 (0.12)	0 (0)	0.53 (0.01)	36	0.99	1123
		50	-3.38 (0.06)	0.20 (0.06)	2.37 (0.07)	4.50 (0.12)	0 (0)	0.50 (0.01)	36	0.99	1087
		60	-3.36 (0.06)	0.20 (0.06)	2.31 (0.06)	4.21 (0.11)	0 (0)	0.49 (0.01)	36	0.99	1013
3	[Ch^+][Cl^-]: 4 TEG	30	-3.21 (0.06)	0.12 (0.06)	2.46 (0.08)	4.87 (0.12)	0 (0)	0.61 (0.02)	37	0.99	1071
		40	-3.21 (0.06)	0.13 (0.06)	2.36 (0.07)	4.59 (0.11)	0 (0)	0.57 (0.01)	37	0.99	1114
		50	-3.22 (0.06)	0.11 (0.06)	2.29 (0.08)	4.33 (0.12)	0 (0)	0.54 (0.02)	37	0.99	900
		60	-3.29 (0.06)	0.13 (0.06)	2.24 (0.07)	4.17 (0.11)	0 (0)	0.51 (0.01)	37	0.99	1045
4	[Ch^+][Cl^-]: 5 TEG	30	-3.39 (0.06)	0.14 (0.07)	2.47 (0.08)	4.90 (0.13)	0.17 (0.10)	0.61 (0.02)	37	0.99	1078
		40	-3.35 (0.06)	0.19 (0.06)	2.37 (0.08)	4.60 (0.12)	0.17 (0.10)	0.56 (0.02)	37	0.99	1099
		50	-3.42 (0.06)	0.16 (0.06)	2.33 (0.07)	4.44 (0.12)	0.15 (0.10)	0.53 (0.01)	37	0.99	1165
		60	-3.45 (0.06)	0.19 (0.06)	2.25 (0.07)	4.15 (0.12)	0.17 (0.09)	0.49 (0.01)	37	0.99	959
5	[Ch^+][Cl^-]: 6 TEG	30	-3.54 (0.06)	0.15 (0.06)	2.44 (0.07)	4.85 (0.11)	0.17 (0.10)	0.61 (0.01)	37	0.99	1095
		40	-3.51 (0.06)	0.17 (0.06)	2.33 (0.07)	4.54 (0.12)	0.18 (0.10)	0.57 (0.01)	37	0.99	1148
		50	-3.42 (0.06)	0.14 (0.06)	2.30 (0.07)	4.29 (0.12)	0.19 (0.09)	0.52 (0.01)	37	0.99	1173
		60	-3.49 (0.06)	0.12 (0.06)	2.20 (0.07)	4.10 (0.11)	0.18 (0.09)	0.50 (0.01)	37	0.99	1239
6	[Ch^+][Cl^-]: 10 TEG	30	-3.31 (0.06)	0.14 (0.06)	2.42 (0.08)	4.82 (0.13)	0.17 (0.10)	0.62 (0.02)	37	0.99	1134
		40	-3.29 (0.06)	0.16 (0.06)	2.31 (0.07)	4.49 (0.12)	0.19 (0.10)	0.58 (0.01)	37	0.99	1194
		50	-3.31 (0.05)	0.15 (0.06)	2.24 (0.07)	4.23 (0.11)	0.20 (0.09)	0.54 (0.01)	37	0.99	1195
		60	-3.36 (0.05)	0.14 (0.06)	2.17 (0.07)	4.02 (0.11)	0.19 (0.09)	0.51 (0.01)	37	0.99	1140
		30	-3.56 (0.06)	0 (0)	2.49 (0.07)	5.17 (0.11)	0 (0)	0.58 (0.01)	35	0.99	958
7	[Ch^+][Ace^-]: 2 TEG	40	-3.62 (0.06)	0 (0)	2.42 (0.07)	5.03 (0.11)	0 (0)	0.53 (0.01)	35	0.99	971
		50	-3.52 (0.06)	0 (0)	2.37 (0.07)	4.87 (0.11)	0 (0)	0.51 (0.01)	36	0.99	1072
		60	-3.54 (0.06)	0 (0)	2.39 (0.07)	4.68 (0.12)	0 (0)	0.50 (0.01)	35	0.99	944
		30	-3.21 (0.06)	0 (0)	2.46 (0.08)	5.06 (0.12)	0 (0)	0.61 (0.02)	35	0.99	1071
8	[Ch^+][Ace^-]: 3 TEG	40	-3.22 (0.06)	0 (0)	2.39 (0.07)	4.81 (0.11)	0 (0)	0.56 (0.01)	35	0.99	1112
		50	-3.19 (0.05)	0 (0)	2.28 (0.06)	4.50 (0.10)	0 (0)	0.52 (0.01)	35	0.99	1303
		60	-3.39 (0.05)	0 (0)	2.31 (0.07)	4.39 (0.10)	0 (0)	0.51 (0.01)	35	0.99	1014
		30	-3.08 (0.07)	0 (0)	2.25 (0.09)	5.13 (0.13)	0 (0)	0.70 (0.01)	34	0.99	882
9	[Ch^+][Ace^-]: 4 TEG	40	-3.09 (0.06)	0 (0)	2.18 (0.07)	4.84 (0.11)	0 (0)	0.65 (0.01)	35	0.99	1119
		50	-3.11 (0.05)	0 (0)	2.12 (0.07)	4.59 (0.10)	0 (0)	0.61 (0.01)	35	0.99	1203
		60	-3.17 (0.05)	0 (0)	2.05 (0.07)	4.30 (0.10)	0 (0)	0.58 (0.01)	36	0.99	1033
		30	-3.25 (0.06)	0 (0)	2.48 (0.07)	5.06 (0.10)	0 (0)	0.62 (0.02)	35	0.99	1417
10	[Ch^+][Ace^-]: 5 TEG	40	-3.41 (0.05)	0 (0)	2.46 (0.07)	4.85 (0.10)	0 (0)	0.60 (0.01)	35	0.99	1481
		50	-3.45 (0.05)	0 (0)	2.39 (0.06)	4.64 (0.10)	0 (0)	0.56 (0.01)	35	0.99	1458
		60	-3.41 (0.05)	0 (0)	2.28 (0.06)	4.32 (0.09)	0 (0)	0.52 (0.01)	35	0.99	1464
		30	-3.38 (0.06)	0 (0)	2.49 (0.08)	5.00 (0.12)	0 (0)	0.63 (0.02)	35	0.99	1214
11	[Ch^+][Ace^-]: 6 TEG	40	-3.38 (0.06)	0 (0)	2.41 (0.07)	4.72 (0.11)	0 (0)	0.59 (0.01)	35	0.99	1352
		50	-3.45 (0.05)	0 (0)	2.35 (0.06)	4.50 (0.10)	0 (0)	0.56 (0.01)	35	0.99	1393
		60	-3.39 (0.05)	0 (0)	2.25 (0.06)	4.18 (0.09)	0 (0)	0.52 (0.01)	35	0.99	1299
		30	-3.32 (0.06)	0 (0)	2.22 (0.07)	4.79 (0.11)	0 (0)	0.62 (0.02)	35	0.99	1358
12	[Ch^+][Ace^-]: 10 TEG	40	-3.28 (0.05)	0 (0)	2.16 (0.07)	4.58 (0.11)	0 (0)	0.61 (0.02)	35	0.99	1371
		50	-3.39 (0.05)	0 (0)	2.11 (0.06)	4.37 (0.10)	0 (0)	0.56 (0.01)	35	0.99	1259
		60	-3.33 (0.06)	0 (0)	2.01 (0.07)	4.16 (0.11)	0 (0)	0.53 (0.01)	35	0.99	1382
		30	-3.22 (0.06)	0.10 (0.06)	2.25 (0.08)	4.57 (0.13)	0.23 (0.10)	0.66 (0.02)	37	0.99	1136
13	TEG	40	-3.20 (0.06)	0.09 (0.06)	2.15 (0.07)	4.29 (0.12)	0.20 (0.10)	0.62 (0.01)	37	0.99	1155
		50	-3.24 (0.05)	0.09 (0.05)	2.09 (0.07)	4.06 (0.11)	0.22 (0.09)	0.59 (0.01)	37	0.99	1310
		60	-3.22 (0.05)	0.09 (0.05)	1.98 (0.07)	3.72 (0.11)	0.23 (0.09)	0.55 (0.01)	37	0.99	1134

[a] *n*, number of probes subjected to multiple linear regression analysis.
[b] R^2, correlation coefficient.
[c] *F*, Fisher F-statistic.

ratios of HBD results in reduced interactions between the chloride anion and the phenolic compounds [39].

3.2. Comparison of DESs with common HBDs

In DES systems 1 and 2, only the anion of the HBA has been altered. Consequently, these were chosen to demonstrate the effect on DES solvation characteristics when the type of choline salt employed as the HBA is varied. Since acetate is an organic anion, stronger dispersive-type interactions (Table 2) were obtained for DES system 2 comprised of [Ch^+][Ace^-] as the HBA compared to DES system 1 composed of the [Ch^+][Cl^-] salt. For example, the *l*-term value of the [Ch^+][Ace^-]: 4 TEG DES was (0.58 ± 0.01) at 60 °C compared to (0.51 ± 0.01) for the [Ch^+][Cl^-]: 4 TEG DES. Acetate also resulted in higher hydrogen bond basicity for DES system 2 compared to DES system 1 as observed from a magnitude of (4.68±0.12) for the [Ch^+][Ace^-]: 2 TEG DES compared to (4.39±0.13) for the [Ch^+][Cl^-]: 2 TEG DES. In addition, DES system 1 generally exhibited stronger dipolar interactions compared to DES system 2, demonstrating a key difference between the solvation characteristics of choline chloride and acetate-based DESs.

A mixture of 16 analytes was chromatographically separated on the [Ch^+][Cl^-]: 4 TEG and [Ch^+][Ace^-]: 4 TEG DESs at 60 °C, as shown in Fig. 2A and B. It can be observed that acidic ana-

Table 3
System constants for neat 1,8-octanediol and deep eutectic solvents comprised of choline chloride as hydrogen bond acceptor and 1,8-octanediol hydrogen bond donor.

No.	DES/neat HBD	Temp. (°C)	System constants								
			c	*e*	*s*	*a*	*b*	*l*	*n* [a]	R^2 [b]	*F* [c]
14	[Ch⁺][Cl⁻]: 2 (1,8-OD)	60	-3.62 (0.07)	-0.07 (0.05)	1.84 (0.07)	4.50 (0.13)	0.33 (0.09)	0.68 (0.01)	36	0.99	870
		70	-3.81 (0.07)	-0.06 (0.06)	1.89 (0.07)	4.46 (0.13)	0.31 (0.09)	0.67 (0.01)	36	0.99	858
15	[Ch⁺][Cl⁻]: 3 (1,8-OD)	60	-3.29 (0.05)	-0.05 (0.05)	1.71 (0.06)	4.18 (0.09)	0.32 (0.08)	0.64 (0.01)	38	0.99	1557
		70	-3.26 (0.04)	-0.03 (0.05)	1.68 (0.06)	3.92 (0.09)	0.30 (0.07)	0.59 (0.01)	38	0.99	1434
16	[Ch⁺][Cl⁻]: 4 (1,8-OD)	60	-3.15 (0.05)	-0.06 (0.05)	1.64 (0.06)	4.00 (0.10)	0.37 (0.08)	0.64 (0.01)	38	0.99	1232
		70	-3.16 (0.04)	-0.05 (0.05)	1.61 (0.06)	3.78 (0.09)	0.33 (0.08)	0.60 (0.01)	38	0.99	1362
17	[Ch⁺][Cl⁻]: 5 (1,8-OD)	60	-3.26 (0.05)	-0.08 (0.05)	1.63 (0.06)	4.03 (0.10)	0.40 (0.08)	0.66 (0.01)	38	0.99	1401
		70	-3.31 (0.05)	-0.05 (0.05)	1.60 (0.06)	3.82 (0.10)	0.39 (0.08)	0.62 (0.01)	38	0.99	1189
18	[Ch⁺][Cl⁻]: 6 (1,8-OD)	60	-3.28 (0.05)	-0.07 (0.05)	1.63 (0.06)	4.07 (0.10)	0.43 (0.08)	0.69 (0.01)	38	0.99	1441
		70	-3.31 (0.05)	-0.08 (0.05)	1.59 (0.06)	3.83 (0.10)	0.44 (0.08)	0.64 (0.01)	38	0.99	1284
19	[Ch⁺][Cl⁻]: 10 (1,8-OD)	60	-3.25 (0.06)	-0.10 (0.06)	1.50 (0.07)	3.89 (0.12)	0.48 (0.10)	0.69 (0.01)	38	0.99	891
		70	-3.31 (0.05)	-0.08 (0.05)	1.48 (0.06)	3.83 (0.10)	0.45 (0.08)	0.65 (0.01)	38	0.99	1254
20	1,8-OD	60	-3.25 (0.07)	0 (0)	1.22 (0.09)	3.49 (0.15)	0.90 (0.12)	0.71 (0.01)	37	0.99	723
		70	-3.27 (0.06)	0 (0)	1.19 (0.08)	3.22 (0.14)	0.87 (0.11)	0.67 (0.01)	37	0.99	801

[a] *n*, number of probes subjected to multiple linear regression analysis.
[b] R^2, correlation coefficient.
[c] *F*, Fisher F-statistic.

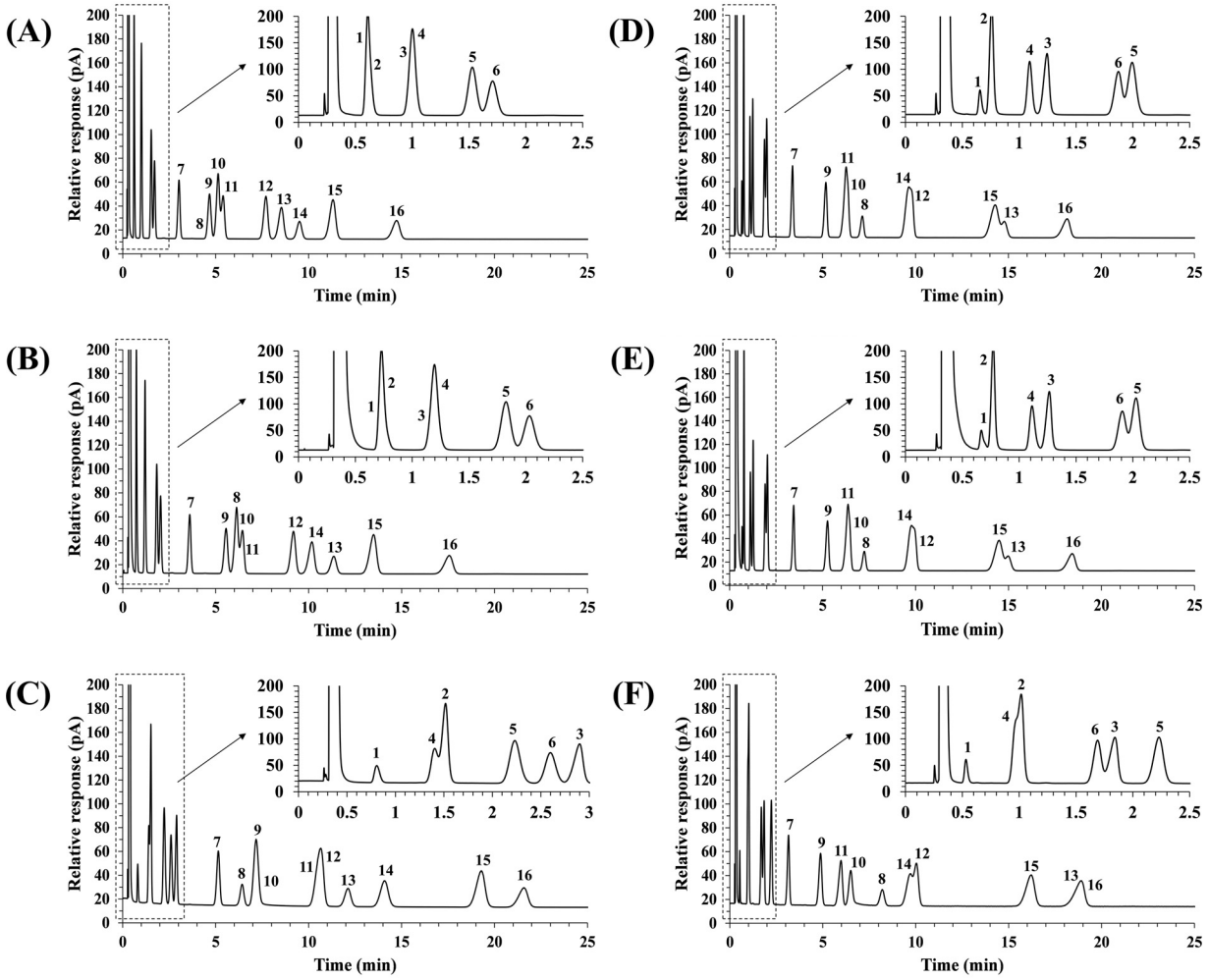


Fig. 1. Chromatographic separation of 16 probe molecules showing the effect of varied HBA ([Ch⁺][Ace⁻]) to HBD (TEG) ratios for DES stationary phases at 50 °C: (A) [Ch⁺][Ace⁻]: 2 TEG, (B) [Ch⁺][Ace⁻]: 3 TEG, (C) [Ch⁺][Ace⁻]: 4 TEG, (D) [Ch⁺][Ace⁻]: 6 TEG, (E) [Ch⁺][Ace⁻]: 10 TEG, and (F) TEG. Analytes: 1, methanol; 2, 1-chlorooctane; 3, 1-bromooctane; 4, 1-propanol; 5, cyclohexanone; 6, 1-butanol; 7, 1-pentanol; 8, *N,N*-dimethylformamide (*N,N*-DMF); 9, cyclopentanol; 10, benzaldehyde; 11, 1-hexanol 12, benzonitrile; 13, *N,N*-dimethylacetamide (*N,N*-DMAC); 14, cyclohexanol; 15, naphthalene; and 16, nitrobenzene.

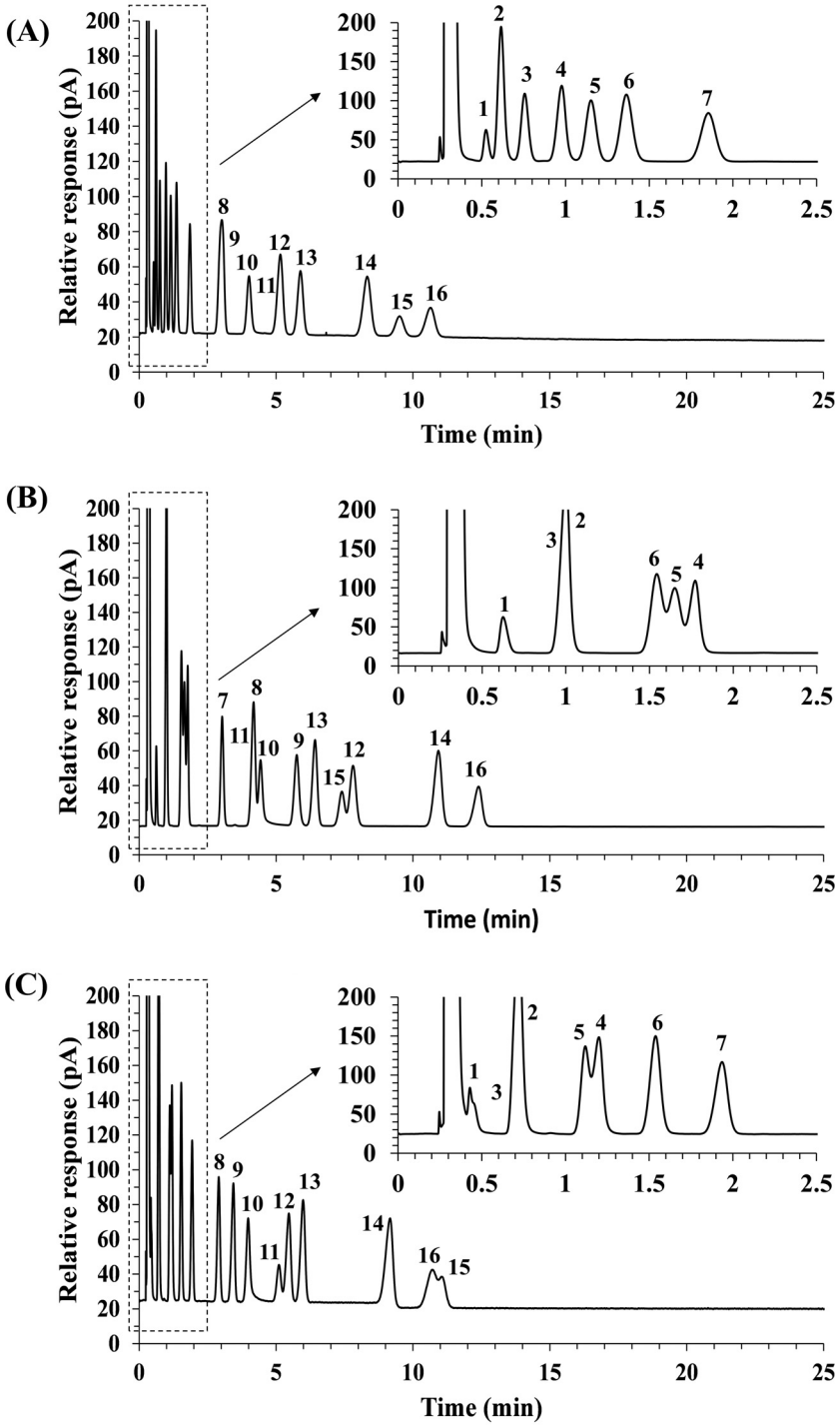


Fig. 2. Separation of a mixture of 16 probe molecules on the (A) [Ch⁺][Cl⁻]: 4 TEG DES, (B) [Ch⁺][Ace⁻]: 4 TEG DES and (C) neat TEG stationary phases at 60 °C. Analytes: 1, methanol; 2, 1-chlorooctane; 3, 1-propanol; 4, 1-bromooctane; 5, 1-butanol; 6, cyclohexanone; 7, 1-pentanol; 8, cyclopentanol; 9, 1-hexanol; 10, benzaldehyde; 11, *N,N*-dimethylformamide (*N,N*-DMF); 12, cyclohexanol; 13, benzonitrile; 14, naphthalene; 15, *N,N*-dimethylacetamide (*N,N*-DMAC); and 16, nitrobenzene.

lytes retain less on DES system 1 compared to DES system 2, further demonstrating the higher hydrogen bond basicity character in acetate-based DESs. For example, 1-pentanol (analyte 7) eluted under 2 min in Fig. 2A (within the inset) compared to over 3 min in Fig. 2B. In contrast, *N,N*-DMAC (analyte 15) was observed to retain over 9 min on the [Ch⁺][Cl⁻]: 4 TEG DES compared to under 8 min on the [Ch⁺][Ace⁻]: 4 TEG DES. The analytes produced excellent peak symmetries but were generally well resolved on the [Ch⁺][Cl⁻]: 4 TEG DES compared to the [Ch⁺][Ace⁻]: 4 TEG DES. The first 7 probe molecules were nearly baseline resolved in Fig. 2A. However, 1-chlorooctane and 1-propanol co-eluted while 1-bromooctane, 1-butanol, and cyclohexanone were poorly resolved from each other in Fig. 2B. Despite having the same HBA/HBD ratio, the analyte elution order was observed to be very different, thereby exemplifying the role of the anion in modifying the solvation properties of DESs comprised of different choline salts.

Table 4
Retention factors of various probe molecules at 60 °C measured on deep eutectic solvents comprised of choline chloride and choline acetate hydrogen bond acceptors.

Probe	DESs/neat HBDs														
	TEG	1,8-OD	[Ch$^+$][Cl$^-$]: 2 TEG	[Ch$^+$][Cl$^-$]: 3 TEG	[Ch$^+$][Cl$^-$]: 4 TEG	[Ch$^+$][Cl$^-$]: 6 TEG	[Ch$^+$][Cl$^-$]: 10 TEG	[Ch$^+$][Ace$^-$]: 2 TEG	[Ch$^+$][Ace$^-$]: 3 TEG	[Ch$^+$][Ace$^-$]: 4 TEG	[Ch$^+$][Ace$^-$]: 6 TEG	[Ch$^+$][Ace$^-$]: 10 TEG	[Ch$^+$][Cl$^-$]: 2 (1,8-OD)	[Ch$^+$][Cl$^-$]: 3 (1,8-OD)	[Ch$^+$][Cl$^-$]: 4 (1,8-OD)	[Ch$^+$][Cl$^-$]: 6 (1,8-OD)	[Ch$^+$][Cl$^-$]: 10 (1,8-OD)
Methanol	0.68	0.72	1.01	0.94	0.87	0.84	0.71	1.11	1.12	1.31	0.91	0.97	0.74	0.93	1.04	0.93	0.82
Ethanol	0.92	1.24	1.01	1.06	0.97	0.93	0.89	1.06	1.19	1.43	1.01	1.11	1.09	1.30	1.48	1.37	1.30
1-Propanol	1.71	2.69	1.60	1.63	1.69	1.62	1.56	1.82	1.91	2.62	1.87	1.97	2.29	2.61	2.98	2.83	2.73
1-Butanol	3.43	6.30	2.71	2.58	3.35	3.27	3.01	3.20	3.75	4.14	3.45	3.53	4.99	5.64	6.48	6.25	6.20
1-Pentanol	6.64	15.14	4.81	5.16	6.01	5.84	5.67	5.73	6.81	10.37	6.65	6.62	10.94	12.49	13.92	14.48	14.15
Cyclopentanol	10.65	19.29	8.42	8.03	10.26	9.81	9.27	9.17	10.86	14.71	10.54	10.73	15.10	17.07	18.90	19.45	18.78
1-Hexanol	13.25	32.27	8.04	8.07	10.77	10.74	10.63	10.37	12.67	20.90	12.60	12.92	23.02	26.03	29.14	30.96	30.56
Cyclohexanol	21.13	43.71	14.52	14.38	18.42	17.98	17.11	16.59	19.82	28.27	19.59	20.51	31.85	35.94	40.05	41.82	41.36
Napthalene	35.88	39.69	21.75	22.32	30.23	29.13	27.66	20.09	26.99	40.01	28.55	29.98	31.20	33.58	39.36	39.92	40.34
Nitrobenzene	42.35	29.39	30.53	30.54	39.20	37.96	36.02	27.88	36.02	45.92	37.47	38.46	33.68	36.43	40.40	39.04	35.71
Acetophenone	35.72	35.87	22.54	23.20	30.01	29.62	28.70	18.96	26.00	32.87	28.66	30.13	29.49	33.03	36.97	38.25	36.77
Benzonitrile	23.41	18.01	16.83	16.66	21.28	20.59	19.58	14.50	19.25	23.39	20.18	20.74	17.60	19.42	21.44	21.12	19.47
Benzaldehyde	14.87	12.97	11.55	11.16	14.27	13.85	13.14	9.38	12.47	15.88	13.55	13.84	12.23	13.80	15.08	14.93	13.70
N,N-Dimethylformamide (*N,N*-DMF)	19.35	21.99	13.71	14.00	14.38	15.76	16.85	9.12	13.07	14.33	15.44	16.38	16.24	17.86	20.67	21.08	21.68
N,N-Dimethylacetamide (*N,N*-DMAC)	43.04	74.42	23.14	25.09	28.23	32.59	35.07	15.62	24.17	26.41	30.84	33.75	39.36	44.57	52.62	56.97	62.72

3.3. Comparison of DESs with the same HBA but different HBDs

DES systems 1 and 3 were chosen to compare system constants and retention characteristics of analytes when the choice of HBA is kept constant but the type of HBD is altered. In DES system 1, TEG possesses ether functional groups that are polar substituents and have the potential to act as additional hydrogen bond acceptor. While 1,8-OD in DES system 3 lacks ether functional groups, the length of the alkyl substituents is kept the same as TEG. Consequently, a comparison of these DES systems truly enables the effect of ether substituents in the HBD to be studied. Slightly higher hydrogen bond basicity was observed for DES system 1 comprised of TEG (Table 2) compared to DES system 3 composed of 1,8-OD as the HBD (Table 3). In contrast, the *l*-term values were generally observed to be smaller for DES system 1 compared to DES system 3, demonstrating higher dispersive-type interactions for DESs comprised of 1,8-OD as the HBD. While *b*-term values were generally zero for DES system 1, strong hydrogen bond acidity of up to (0.48 ± 0.10) was achieved for the [Ch$^+$][Cl$^-$]: 10 (1,8-OD) DES at 60 °C. The retention factors of basic probe molecules (Table 4) provide further evidence of higher hydrogen bond acidity in DES system 3. In addition, DES system 3 generally exhibited lower n-π and π-π interactions compared to DES system 1, as 1,8-OD lacks the lone pair of electrons that are present on ether functional groups in TEG molecules.

3.4. Comparison of DESs with their respective HBDs

DES systems 1 and 2 were directly compared with TEG to illustrate the differences in solvation properties between neat HBDs and their DESs. Since neat HBDs do not possess the chloride anion that acts as the primary hydrogen bond acceptor, the hydrogen bond basicity was determined to be the lowest for neat TEG (3.72 ± 0.11) at 60 °C (Table 2) compared to all eutectic mixtures of DES systems 1 and 2. In contrast, its hydrogen bond acidity of (0.23 ± 0.09) was the highest compared to its DESs. A similar trend was observed for 1,8-OD as its hydrogen bond basicity of (3.49 ± 0.15) strongly contrasted with a minimum of (3.89 ± 0.12) for DES system 3 while its hydrogen bond acidity of (0.90 ± 0.12) was significantly higher compared to a maximum of (0.48 ± 0.10) for its DESs. Both TEG and 1,8-OD generally possessed higher dispersive-type interactions and lower dipolarity compared to their DESs.

A total of 16 probe molecules were separated chromatographically on TEG at 60 °C, as shown in Fig. 2C. By comparing the separation of the same set of probes on the [Ch$^+$][Cl$^-$]: 4 TEG and [Ch$^+$][Ace$^-$]: 4 TEG eutectic mixtures, key differences in analyte retention characteristics between neat TEG and its DESs were obtained. For example, naphthalene (analyte 14) retained for approximately 9 min on TEG compared to over 10.5 min on the [Ch$^+$][Ace$^-$]: 4 TEG DES and under 9 min on the [Ch$^+$][Cl$^-$]: 4 TEG DES. *N,N*-DMAC (analyte 15) was observed to be the highest retaining probe on TEG (approximately 11 min) compared to its retention of under 10 and 8 min on the [Ch$^+$][Cl$^-$]: 4 TEG and [Ch$^+$][Ace$^-$]: 4 TEG DESs, respectively. Similar analyte behavior based on their retention factors (Table 4) can be observed for 1,8-OD and its DESs. Clearly, the chromatographic retention of probe molecules on neat HBDs is very different from their DESs.

3.5. Comparison of system constants and analyte retention of neat HBDs

Neat 1,8-OD and TEG were both compared to determine differences in their solvation characteristics as they can partially influence the solvation properties of their DESs. Since TEG has more polar functional groups (i.e., ether linkages), it was observed to

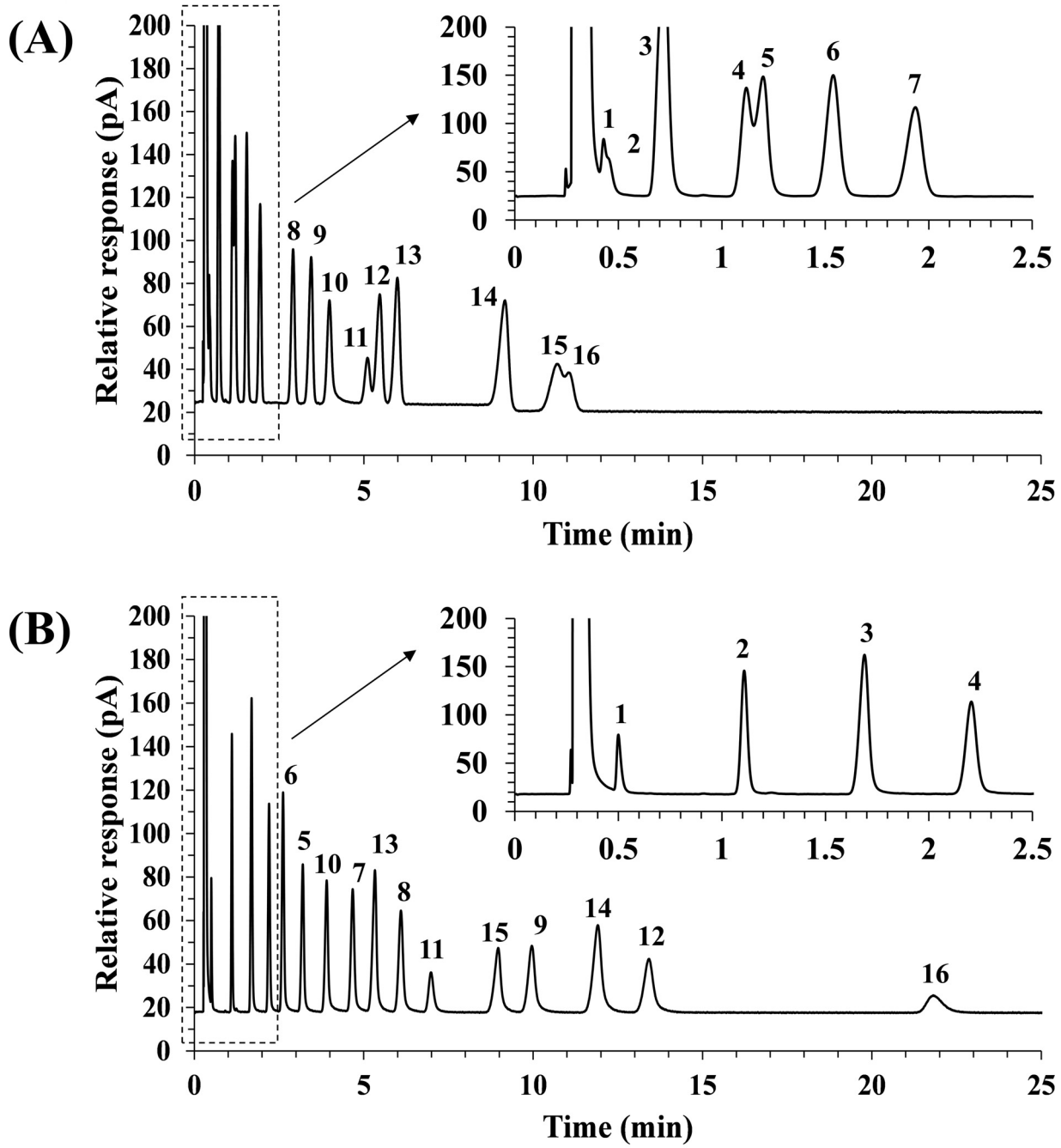


Fig. 3. Comparison of chromatographic separation on neat HBDs: (A) TEG and (B) 1,8-OD stationary phases at 60 °C. Analytes: 1, methanol; 2, 1-propanol; 3, 1-chlorooctane; 4, 1-butanol; 5, 1-bromooctane; 6, cyclohexanone; 7, 1-pentanol; 8, cyclopentanol; 9, 1-hexanol; 10, benzaldehyde; 11, *N,N*-dimethylformamide (*N,N*-DMF); 12, cyclohexanol; 13, benzonitrile; 14, naphthalene; 15, nitrobenzene; and 16, *N,N*-dimethylacetamide (*N,N*-DMAC).

possess lower dispersive-type interactions (0.55 ± 0.01) at 60 °C (Table 2) compared to (0.71 ± 0.01) for 1,8-OD (Table 3). The hydrogen bond basicity of (3.72 ± 0.11) for TEG was slightly higher compared to (3.49 ± 0.15) for 1,8-OD. However, a significantly large difference in hydrogen bond acidity was observed between TEG (0.23 ± 0.09) and 1,8-OD (0.90 ± 0.12). In addition, 1,8-OD possessed lower dipolarity of (1.22 ± 0.09) compared to (1.98 ± 0.07) for TEG.

The chromatographic separation of 16 probe molecules at 60 °C was observed to be very different between neat TEG and 1,8-OD, as shown in Fig. 3. While *N,N*-DMAC (analyte 16) was the highest retaining analyte on both solvents, its retention was approximately twice that of TEG on 1,8-OD. Similarly, naphthalene (analyte 14) retained for approximately 9 min on TEG and around 12 min on 1,8-OD. In addition, haloalkanes (1-chlorooctane and 1-bromooctane) were found to retain longer on 1,8-OD compared to TEG, further exemplifying higher dispersion interactions for neat 1,8-OD. Furthermore, analyte peaks appeared to possess tailing characteristics on 1,8-OD but were generally baseline resolved compared to TEG. Thus, differences between the 2 solvents could be observed in both their analyte retention and system constants.

4. Conclusions

Choline chloride and acetate-based DESs are designer solvents that have been extensively employed in separations, organic syn-

thesis, and catalysis due to their versatile and biodegradable nature. Since pairing choline chloride as the HBA with a specific polyol as HBD has been known to form multiple eutectic mixtures each possessing a unique stoichiometric ratio between the HBA and HBD, most studies often screen all possible eutectic combinations of a DES system to identify the one that possesses the optimal HBA/HBD ratio. While modifying the molar ratio of HBD serves as a useful tool to alter the physical and chemical properties of a DES, its impact on solvation properties has not been fully described using the established solvatochromic dye methods. In contrast, the chromatographic approach employed in this study is capable of thoroughly investigating minor variations in individual solvation interactions of DESs when their HBA/HBD ratio is systematically modified.

For the DES systems explored in this study, variations in hydrogen bond basicity and acidity induced by small changes in HBA/HBD ratio were determined to be more drastic compared to dispersive-type and dipolar interactions. Hydrogen bond basicity was generally reduced while hydrogen bond acidity was significantly enhanced when the molar ratio of HBD was increased. Although the decrease in hydrogen bond basicity generally plateaued at higher molar ratios of HBD, hydrogen bond acidity continued to increase up to molar ratios of 1:10. Chromatographic retention of different classes of analytes was significantly altered when systematic changes were made to the HBA/HBD ratio. While acidic probes generally retained the longest on the eutectic mixture comprised of a 1:4 molar ratio, the retention of basic probes was stronger at higher molar ratios of HBD. It was also observed that DESs comprised of $[Ch^+][Ace^-]$ as the HBA exhibited higher hydrogen bond basicity and dispersion interactions compared to those composed of the $[Ch^+][Cl^-]$ salt. The presence of ether functional groups in TEG generally lead to DESs possessing higher hydrogen bond basicity and n-π and π-π interactions compared to those composed of 1,8-OD as the HBD. It was also established that TEG and 1,8-OD possessed higher hydrogen bond acidity and dispersion interactions and significantly lower hydrogen bond basicity and dipolarity compared to DESs incorporating them as the HBDs. Additionally, the individual solvation interactions of neat TEG and 1,8-OD were observed to be significantly different, further exemplifying the influence that the choice of HBD can have on DES solvation characteristics. This study highlights the key role played by molar ratios of HBA and HBD in regulating the solvation interactions of choline chloride-based DESs and demonstrates that a suboptimal choice of eutectic mixture in any DES system can adversely impact its effectiveness as an extractant, catalyst or reaction medium, thereby urging the need to design and employ DESs on the basis of their unique solvation properties.

Declaration of Competing Interest

The authors declare no conflicts of interest in this work.

CRediT authorship contribution statement

Nabeel Mujtaba Abbasi: Writing – original draft, Writing – review & editing. **Muhammad Qamar Farooq:** Writing – review & editing. **Jared L. Anderson:** Supervision, Writing – review & editing, Project administration, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.chroma.2022.462871.

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